



Rajshahi University of Engineering & Technology (RUET)

Department of Electrical & Computer Engineering

Lab Report - 1

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Comprehensive Study of Error Measurements in Numerical Methods

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1 Introduction to Numerical Errors

Numerical analysis is the study of algorithms that use numerical approximation for the problems of continuous mathematics. Unlike analytical methods, which yield exact solutions, numerical methods rely on approximations. Because exact mathematical solutions are often difficult, computationally expensive, or impossible to obtain, these numerical techniques are essential in engineering and computer science.

However, the reliance on approximation introduces a fundamental concept: **Error**. The difference between the exact (true) solution and the computed approximate solution is defined as the error. Understanding, quantifying, bounding, and minimizing these errors is the most critical aspect of numerical analysis. If errors are not properly managed, they can compound and render the final computed result entirely meaningless. This report provides a detailed mathematical exploration of the various types of errors, how they are measured, and how they propagate through calculations.

2 Required Tools

- **Hardware:** Mac M4 Air (For mathematical computation and typesetting).
- **Software/Environment:** LaTeX via Overleaf (For structured report generation and mathematical typesetting).
- **Analytical Tools:** Standard algebraic expansion and calculus theorems for manual error verification.

3 Significant Figures and Precision

Before detailing the types of errors, it is necessary to establish the language used to describe numerical precision: significant figures.

3.1 Definition of Significant Figures

Significant figures (or digits) are the reliable digits in a number that are known with certainty, plus one estimated (uncertain) digit. The concept of significant digits is closely tied to how errors are bounded in numerical calculations.

Rules for determining significant figures:

1. All non-zero digits are significant. (e.g., 1.234 has four significant digits).
2. Zeros between non-zero digits are significant. (e.g., 1002 has four significant digits).
3. Leading zeros are never significant; they merely indicate the position of the decimal point. (e.g., 0.00045 has two significant digits).

4. Trailing zeros in a number containing a decimal point are significant. (e.g., 45.00 has four significant digits).

3.2 Accuracy vs. Precision

While often used interchangeably in casual conversation, these terms have distinct mathematical meanings in numerical methods:

- **Accuracy:** Refers to how closely a computed or measured value agrees with the true (exact) value.
- **Precision:** Refers to how closely individual computed or measured values agree with each other, or the number of significant figures used to represent the value.

A numerical method can be highly precise (generating results to 10 decimal places) but entirely inaccurate (if the underlying algorithm is flawed or unstable).

4 Types of Errors in Numerical Methods

Errors in numerical computations generally stem from three primary classifications: Inherent errors, Round-off errors, and Truncation errors.

4.1 Inherent Errors

Inherent errors are those that are already present in the initial data before any numerical computation even begins. These can be sub-divided into:

- **Data Errors:** Errors arising from physical measurements. No measuring instrument is perfectly precise.
- **Conversion Errors:** Errors occurring when converting physical constants into a machine's limited representation.
- **Human Errors:** Blunders in reading data or entering parameters.

Mathematical Example: Consider calculating the area of a circle $A = \pi r^2$. If the radius r is measured with a ruler that only has millimeter markings, a true radius of $r = 5.236$ cm might be inherently recorded as $r = 5.2$ cm. The true area is $A_{true} = \pi(5.236)^2 \approx 86.126$ cm². The computed area based on the inherent error is $A_{computed} = \pi(5.2)^2 \approx 84.949$ cm². The inherent error in measurement cascaded into an error of ≈ 1.177 cm² in the final result.

4.2 Round-off Errors

Round-off errors occur because computing devices can only represent numbers using a finite number of bits. When a number with infinite digits (like π or $1/3$) or simply more digits than the computer can store is processed, it must be approximated.

4.2.1 Floating-Point Representation

Computers store numbers in normalized floating-point form:

$$fl(x) = \pm m \times \beta^e$$

Where:

- m is the mantissa (fractional part), such that $\frac{1}{\beta} \leq m < 1$.
- β is the base (usually 2 for binary, or 10 for decimal examples).
- e is the exponent, bounded by $L \leq e \leq U$.

If a computer can only store d digits for the mantissa, a number must be either **chopped** or **rounded**.

Example of Chopping vs. Rounding: Suppose a computer operates in base-10 ($\beta = 10$) and stores exactly 4 significant digits in its mantissa. Let the true number be $x = 0.654381 \times 10^2$.

- **Chopping:** Simply discard all digits beyond the 4th digit.
 $fl_{chop}(x) = 0.6543 \times 10^2 = 65.43$
- **Symmetric Rounding:** Look at the 5th digit. If it is ≥ 5 , add 1 to the 4th digit. Here, the 5th digit is 8.
 $fl_{round}(x) = 0.6544 \times 10^2 = 65.44$

4.2.2 Machine Epsilon (ϵ)

Machine epsilon is the maximum relative error introduced by round-off. It is defined as the smallest positive number ϵ such that:

$$fl(1 + \epsilon) > 1$$

For a system using base β and t digits of precision:

$$\begin{aligned}\epsilon_{chop} &= \beta^{1-t} \\ \epsilon_{round} &= \frac{1}{2}\beta^{1-t}\end{aligned}$$

Any numerical algorithm must ensure that errors do not amplify machine epsilon disproportionately.

4.3 Truncation Errors

Truncation errors arise when an infinite mathematical process is arbitrarily stopped (truncated) to form a finite process. This is most commonly seen when using series expansions, such as Taylor or Maclaurin series, to approximate transcendental functions (e.g., e^x , $\sin(x)$, $\cos(x)$), or in numerical integration.

4.3.1 Taylor Series Expansion and Remainder

The Taylor series expansion of a continuously differentiable function $f(x)$ around a base point a is given by:

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n + R_n(x)$$

Where $R_n(x)$ is the Remainder term, representing the exact **truncation error** introduced by stopping the series after n terms. The Lagrange form of the remainder is:

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - a)^{n+1}$$

for some ξ between a and x .

4.3.2 Extensive Mathematical Example: Truncating e^x

Consider evaluating $f(x) = e^x$ at $x = 0.5$ using a Maclaurin series (Taylor series centered at $a = 0$). The true value (to 6 decimal places) is:

$$e^{0.5} \approx 1.648721$$

The Maclaurin series for e^x is:

$$e^x \approx 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$

Let us calculate the approximation and the corresponding truncation error by varying the number of terms (N).

Terms (N)	Polynomial $P_N(0.5)$	Approximate Value	Truncation Error $ e^{0.5} - P_N $
1	1	1.000000	0.648721
2	$1 + 0.5$	1.500000	0.148721
3	$1.5 + \frac{0.5^2}{2}$	1.625000	0.023721
4	$1.625 + \frac{0.5^3}{6}$	1.645833	0.002888
5	$1.645833 + \frac{0.5^4}{24}$	1.648438	0.000283

Table 1: Truncation Error in Maclaurin Series Approximation of $e^{0.5}$

Analysis of the Remainder: If we stop at $n = 3$ terms (Polynomial $N = 4$), the truncation error is precisely given by the remainder formula:

$$R_3(0.5) = \frac{e^\xi}{4!}(0.5)^4$$

where $0 < \xi < 0.5$. The maximum possible error occurs if $\xi = 0.5$.

$$R_3(0.5) \leq \frac{e^{0.5}}{24} \times 0.0625 = \frac{1.648721}{24} \times 0.0625 \approx 0.00429$$

As seen in the table, the actual error 0.002888 is strictly bounded by the theoretical maximum truncation error 0.00429.

5 Error Measurement Formulas

To evaluate the magnitude and practical impact of the errors discussed above, mathematical metrics must be defined. Let X_T be the true (exact) value and X_A be the approximate (computed) value.

5.1 Absolute Error (E_a)

The absolute error is the physical magnitude of the difference between the true value and the approximate value.

$$E_a = |X_T - X_A|$$

Example: Suppose the exact distance between two buildings is $X_T = 1205.5$ meters, but a surveying algorithm yields $X_A = 1204.0$ meters.

$$E_a = |1205.5 - 1204.0| = 1.5 \text{ meters}$$

While mathematically straightforward, absolute error lacks context. An error of 1.5 meters is insignificant when measuring the distance to the moon, but catastrophically large when designing a microchip.

5.2 Relative Error (E_r)

Relative error normalizes the absolute error by comparing it to the size of the true value. This provides the missing context.

$$E_r = \frac{|X_T - X_A|}{|X_T|} = \frac{E_a}{|X_T|}$$

Example: Using the previous surveying example:

$$E_r = \frac{1.5}{1205.5} \approx 0.001244$$

This shows that the error is roughly one-thousandth of the total measurement, which provides a much better sense of the approximation's quality.

5.3 Percentage Error (E_p)

The percentage error is simply the relative error expressed as a percentage. It is the most universally understood metric in engineering documentation.

$$E_p = E_r \times 100\% = \left(\frac{|X_T - X_A|}{|X_T|} \right) \times 100\%$$

Example: Continuing the surveying calculation:

$$E_p = 0.001244 \times 100\% = 0.1244\%$$

A percentage error of 0.1244% indicates a highly accurate approximation.

6 Propagation of Errors

In numerical methods, an algorithm requires a sequence of mathematical operations. If the initial inputs contain errors (inherent or round-off), those errors will propagate through subsequent additions, multiplications, and divisions. It is crucial to determine if an algorithm is *stable* (errors decay or remain bounded) or *unstable* (errors grow exponentially).

Let variables x and y have exact values, and let their respective absolute errors be Δx and Δy . The approximate values are $x_A = x + \Delta x$ and $y_A = y + \Delta y$.

6.1 Error in Addition and Subtraction

Consider the function $z = x + y$. The approximate result is:

$$z_A = (x + \Delta x) + (y + \Delta y) = (x + y) + (\Delta x + \Delta y) = z + \Delta z$$

Thus, the absolute error in addition is bounded by the sum of absolute errors:

$$|\Delta z| \leq |\Delta x| + |\Delta y|$$

This implies that absolute errors compound linearly during addition.

6.2 Error in Multiplication

Consider $z = x \times y$.

$$z_A = (x + \Delta x)(y + \Delta y) = xy + x\Delta y + y\Delta x + \Delta x\Delta y$$

Assuming Δx and Δy are small, their product $\Delta x\Delta y$ is negligible.

$$z_A - z = \Delta z \approx x\Delta y + y\Delta x$$

Dividing by the true value $z = xy$:

$$\frac{\Delta z}{z} \approx \frac{x\Delta y}{xy} + \frac{y\Delta x}{xy} = \frac{\Delta y}{y} + \frac{\Delta x}{x}$$

Thus, for multiplication, the **relative errors** are additive:

$$E_{r,z} \leq E_{r,x} + E_{r,y}$$

6.3 Error in Division

Consider $z = \frac{x}{y}$. Using Taylor series approximation for multivariable functions, it can similarly be shown that relative errors are also additive in division:

$$\frac{\Delta z}{z} \approx \frac{\Delta x}{x} - \frac{\Delta y}{y} \implies \left| \frac{\Delta z}{z} \right| \leq \left| \frac{\Delta x}{x} \right| + \left| \frac{\Delta y}{y} \right|$$

7 Loss of Significance (Subtractive Cancellation)

One of the most dangerous phenomena in numerical computing is **Loss of Significance**, also known as subtractive cancellation. This occurs when two nearly equal numbers are subtracted, causing a massive increase in the relative error and destroying the significant figures of the result.

7.1 Mathematical Example of Subtractive Cancellation

Consider the calculation of $z = x - y$. Let $x = 0.1234567$ and $y = 0.1234566$. Assume our computer can only store 6 significant digits (rounding the 7th digit). The true values

rounded to 6 digits are:

$$x_A = 0.123457$$

$$y_A = 0.123457$$

The exact calculation should be:

$$z_{true} = 0.1234567 - 0.1234566 = 0.0000001 = 1.0 \times 10^{-7}$$

However, the computer evaluates the rounded inputs:

$$z_{computed} = x_A - y_A = 0.123457 - 0.123457 = 0.0$$

The absolute error is:

$$E_a = |1.0 \times 10^{-7} - 0.0| = 1.0 \times 10^{-7}$$

The relative error is catastrophic:

$$E_r = \frac{1.0 \times 10^{-7}}{1.0 \times 10^{-7}} = 1.0 = 100\% \text{ Error}$$

Because the numbers were so close, the subtraction destroyed all meaningful data, resulting in a 100% relative error.

7.2 Algorithmic Mitigation

To fix subtractive cancellation, numerical analysts rewrite the algebraic expression to avoid subtracting nearly equal numbers. Consider evaluating the function $f(x) = \sqrt{x+1} - \sqrt{x}$ for a very large x , say $x = 10^5$. $\sqrt{100001} \approx 316.2293$ and $\sqrt{100000} \approx 316.2278$. Subtraction causes cancellation.

By rationalizing the numerator:

$$f(x) = (\sqrt{x+1} - \sqrt{x}) \times \frac{\sqrt{x+1} + \sqrt{x}}{\sqrt{x+1} + \sqrt{x}} = \frac{(x+1) - x}{\sqrt{x+1} + \sqrt{x}} = \frac{1}{\sqrt{x+1} + \sqrt{x}}$$

Now, for $x = 10^5$:

$$f(x) = \frac{1}{316.2293 + 316.2278} = \frac{1}{632.4571} \approx 0.00158113$$

This alternative formulation involves addition in the denominator rather than subtraction in the numerator, entirely avoiding the loss of significance and preserving full machine precision.

8 Discussion

Through rigorous mathematical analysis, it becomes evident that numerical error is not a singular flaw but a multifaceted property of digital computation.

- We observed that **truncation errors** are a mathematically deterministic consequence of approximation schemes (like the Taylor series) and can be minimized simply by increasing the order of approximation (i.e., adding more terms).
- Conversely, **round-off errors** are a hardware and data-structure limitation bound by Machine Epsilon.
- The propagation analysis highlighted a fundamental theorem of numerical methods: while additions accumulate absolute error, multiplications accumulate relative error.
- The most critical finding was the danger of subtractive cancellation, demonstrating that an algebraically correct formula is not necessarily a computationally stable one.

9 Conclusion

The comprehensive study of error measurements confirms that numerical analysis is essentially the mathematics of managing uncertainty. By carefully defining absolute, relative, and percentage errors, we establish criteria for algorithmic termination and validity. Understanding how truncation and round-off errors combine allows engineers and computer scientists to design stable algorithms, rearrange algebraic formulas to prevent loss of significance, and ensure that computed results safely reflect physical reality.

References

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